

PRODUCT MANAGEMENT CASE STUDY

TripKhat a

How I turned a real Shirdi trip expense argument into a mobile-first group expense tracker — and the product thinking behind every decision.

Fintech · Social | Mobile-First | 0→1 Product | India Market | Solo PM
Build

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01 INTRODUCTION

A product born from a real *argument at a dhaba.*

TripKhata started not in a boardroom or a hackathon — it started on a trip to **Shirdi** with six friends and one unanswered question: **"Exactly who owes how much to whom?"**

What followed was forty minutes of cross-referencing WhatsApp messages, Notes app screenshots, and fading memory — a ritual I later discovered was universal to every Indian friend-group trip. TripKhata is my attempt to eliminate that ritual entirely.

This case study documents the full product thinking journey: from identifying the problem, validating the gap, defining the user, and making hard prioritization tradeoffs — to building an MVP that is live, functional, and being used by real people.

CONTEXT

TripKhata is a progressive web app (PWA) built as a solo project. I wore multiple hats: PM, designer, developer, and first user. Every decision documented here reflects real tradeoffs I faced.

02 PROBLEM STATEMENT

Group travel creates *financial fog* — and no one talks about it.

In most Indian friend-group trips, expenses are spontaneous and distributed. One person pays for the train, another for the hotel, someone handles the auto fares, and nobody tracks the chai-pani or the entry tickets. By day two, the group is operating on estimated memory instead of ground truth.

"The problem isn't the money. The problem is the discomfort of asking friends to pay back — without proof, without transparency."

CORE INSIGHT — USER RESEARCH

This creates three compounding problems: **cognitive overhead** (remembering who paid what), **social friction** (awkward settlement conversations), and **trust erosion** (disputes over amounts, even among close friends). The consequence isn't just inconvenience — it quietly damages relationships.

WHAT USERS CURRENTLY DO

- Notes app / paper lists
- WhatsApp group tracking
- Designated "accountant" friend
- Rough mental math
- Google Sheets (rare)
- Splitwise (too complex)

WHERE EACH METHOD BREAKS

- Notes: no multi-person view
- WhatsApp: scrolling, version chaos
- Accountant: single point of failure
- Mental math: errors accumulate
- Sheets: mobile UX is terrible
- Splitwise: feature overload, not India-first

03 REAL USER PAIN POINTS

Five pain points every *group traveler* recognizes instantly.

Through conversations with students, young professionals, and frequent travelers in Tier 2 and Tier 3 cities, I mapped the following recurring frustrations — verbatim complaints heard repeatedly.



Memory Overload

Tracking 15+ micro-expenses across a 3-day trip is cognitively impossible. Users forget small amounts, and small amounts add up.



Social Awkwardness

"Yaar, tu mujhe 340 de" — asking for exact amounts from friends feels transactional and uncomfortable, especially after a good trip.



Settlement Complexity

In a 6-person group, balancing 15+ bilateral debts manually is error-prone. Most groups overpay or underpay and call it even.



No Offline Capability

Hill stations, pilgrimage sites like Shirdi, beaches — network is unreliable exactly where groups travel. Existing apps fail here.



Too Many Features

Apps like Splitwise offer currency conversion, recurring expenses, and receipt scanning. Traveling students want none of that. Complexity kills adoption.



No Clear Trip End

There's no ritual to "close" a trip and declare everyone settled. Debts linger in apps and in conversations for weeks.

04 MARKET GAP ANALYSIS

Existing solutions solve the *wrong problem* for the wrong user.

The expense-splitting app space is not empty. But a competitive audit reveals a critical misalignment: existing products are designed for Western, urban, high-income users who need roommate or couple expense management — not Indian college students splitting a group Shirdi trip.

MARKET INSIGHT

India has 500M+ smartphone users under 30, with group travel being one of the fastest-growing youth behaviors post-pandemic. Yet no app in the top 10 Play Store results for "group expense" is optimized for short-trip, friend-group, India-first use cases.

App	Target User	Key Weakness	TripKhata Advantage
Splitwise	Global, roommates, couples	Feature-heavy, slow onboarding	Trip-first architecture, zero learning curve
Google Pay / PhonePe	P2P payments	No expense grouping, no split logic	Full expense timeline + settlement suggestions
Tricount	Travel groups globally	Unknown in India, bland UX	India-native, Rs-first design, shareable reports
Excel / Notes	Tech-comfortable users	Manual, no automation, no settlement	Auto-split, auto-settle, PDF report generation

The gap is clear: **simplicity + trip-first UX + India-first design** is entirely unclaimed territory in this space.

05 USER PERSONA

Designed for one person, used by *millions like them.*

Rather than building abstract personas, I built TripKhata for a real archetype I interact with daily: the young Indian who travels in groups, pays using UPI, and communicates entirely on WhatsApp.

Rohan Deshmukh

PRIMARY PERSONA · COLLEGE GROUP TRAVELER

Age 20–26 · Tier 2/3 City · UPI-native · Budget Traveler · Android User · Group of 4–8

GOALS

- Track who paid what without friction
- Settle fairly without awkward asks
- Keep a souvenir record of the trip
- Not be "that guy" who loses count

FRUSTRATIONS

- Existing apps require too many steps
- Friends won't install new apps
- Post-trip settlement always gets messy
- Can't share final summary easily

PM NOTE

I deliberately avoided building for the power user who needs multi-currency or receipt OCR. Serving Rohan exceptionally well creates a product that's genuinely loved — rather than a bloated product that's barely tolerated.

06 PRODUCT VISION

The trip should be remembered for the *experiences, not the expenses.*

TripKhata's vision is to make financial transparency in friend groups effortless — removing the friction of money from the joy of travel. Every product decision is tested against one question: does this make the user's trip easier or just the app more complicated?

WHAT WE BUILD

- Instant expense logging (under 10 seconds)
- Automatic equal splitting
- Minimum-transaction settlement algorithm
- One-tap PDF report to share on WhatsApp
- Offline-first architecture
- No-login, no-friction onboarding

WHAT WE DELIBERATELY AVOID

- Currency conversion
- Receipt scanning / OCR
- Social feed or activity streams
- Bank account linking
- Subscription model (MVP)
- Complex custom split percentages

STRATEGIC PRINCIPLE

In product, what you choose NOT to build is often more important than what you build. Every feature excluded from TripKhata's MVP was a deliberate choice to protect the core experience: radical simplicity.

07 MVP THINKING

The smallest product that *actually solves the problem* — nothing more.

The most common MVP mistake is shipping a half-built version of a big idea. TripKhata's MVP approach was different: **ship a complete solution to a narrow problem**. The problem we chose: a group of 4–6 friends needs to track and settle trip expenses without any prior setup.

MVP SUCCESS DEFINITION

A user with zero training can create a trip, add members, log an expense, and see a clear settlement suggestion — in under 90 seconds, on a mobile device, without creating an account.

This framing eliminated an entire category of features. Account creation, authentication, cloud sync, notifications, and social sharing were all deferred. The initial product lives entirely in local browser storage — zero backend, zero signups, zero drop-off from onboarding friction.

Feature	In MVP?	Reasoning
Create trip + add members	✓ Yes	Core to product identity — must work
Log expense with payer	✓ Yes	Primary job-to-be-done
Auto equal split	✓ Yes	90% of trip expenses split equally
Settlement suggestions	✓ Yes	Core value proposition — what makes app worth it
PDF export / report	✓ Yes	Shareable output = social proof + utility
Expense categories	~ Partial	Basic tags only — full taxonomy deferred
Custom split %	✗ Deferred	Edge case — adds cognitive load for 90% of users

Feature	In MVP?	Reasoning
Cloud sync / accounts	✗ Deferred	Onboarding friction > benefit at MVP stage
UPI integration	✗ Deferred	Regulatory complexity; settlement suggestion is sufficient

08 FEATURE PRIORITIZATION

Every feature is a tradeoff. *Here's how I chose.*

I used a modified **RICE framework** (Reach × Impact × Confidence ÷ Effort) combined with a gut check: *"Would removing this feature break the product's core promise?"* If yes, it ships. If no, it waits.

Feature	Priorit y	Effort	Impact	Decision
Trip creation + member setup	P0	Low	Very High	Ship Now
Expense log + auto-split	P0	Low	Very High	Ship Now
Settlement algorithm	P0	Mediu m	Very High	Ship Now
Expense timeline view	P1	Low	High	Ship Now
PDF trip report	P1	Mediu m	High	Ship Now
Personal expense tracking	P2	Low	Medium	Ship Now
Shared trip wallet	P2	Mediu m	Medium	Ship Now
Custom split (% or Rs)	P3	High	Low-Mediu m	Next Version
User accounts + cloud sync	P3	Very High	Medium	Next Version

Feature	Priorit y	Effort	Impact	Decision
UPI payment integration	P3	Very High	High	Future

PRIORITIZATION INSIGHT

P0 features define whether the product is usable. P1 features define whether it's lovable. P2 features define whether users return. P3+ features define whether it's fundable. I shipped up to P2 in the first version.

09 UX DECISIONS

Every friction point removed is a *relationship preserved.*

The most important UX principle in TripKhata is: **never make the user think more than the problem itself already requires.** Below are the six most consequential UX decisions — and the reasoning behind each.



No Account Required

The highest drop-off in apps happens at signup. TripKhata starts with a trip name and member names — nothing else. You're adding expenses in 30 seconds.



Mobile-Only Viewport

Group expenses are logged in the moment — at the restaurant, at the toll booth, at the Shirdi temple entry. Every interaction is thumb-friendly.



Dark Mode as Default

Most logging happens in low-light environments (night travel, evening chai, early morning at Shirdi). Dark mode reduces friction in the primary use context.



"Mark Settled" Ritual

Settling debt needs a moment of closure. A clear "Mark Settled" button with visual confirmation gives users a satisfying ritual — debt is officially done.



PDF Report as Social Share

The PDF isn't just a receipt — it's a trip memory. Designed to be shareable on WhatsApp, it doubles as a lightweight viral mechanism for word-of-mouth growth.



Minimum Transfers Algorithm

In a 6-person group, naive settlement requires 15 transfers. TripKhata's algorithm reduces this to 3–4. Fewer transfers = less awkwardness.

10 USER FLOW

From "we're going on a trip" to *"we're all settled"* — in 6 steps.

The entire user journey was mapped to minimize cognitive load at each step. Every screen has one primary action. There are no dead ends.

Create Trip

1

User enters trip name (e.g., "Shirdi June Trip"). No account, no password. One field. Done.

Add Members

2

Add friend names with emoji avatars. Quick, visual, no contact list permission required. Typically done in 60 seconds.

Log Expenses (Ongoing)**3**

One-tap FAB button → expense title → amount → who paid → who's included. Auto-split calculated instantly. Takes under 10 seconds per expense.

Review Timeline**4**

Chronological expense feed gives the group a shared, transparent record. Anyone can check who paid what, when.

Check Settlement Tab**5**

App shows the minimum number of transfers needed. Clear "A pays B → Rs X" format. No ambiguity, no arguments.

Mark Settled + Export PDF**6**

One-tap settlement confirmation. Trip closes. PDF report generated for WhatsApp sharing. Closure achieved — financially and emotionally.

DESIGN PRINCIPLE

Every step in the user flow has exactly one primary action. When users know what to do next without thinking, they do it. Confused users abandon products — not their trips.

11 KEY CHALLENGES

The hard problems that shaped
every product decision.

TECHNICAL CHALLENGES

- Minimum-transfer settlement algorithm without negative balances
- Offline-first data persistence without backend
- PDF generation client-side without server
- Floating point errors in multi-person splits
- PWA install experience across Android launchers

PRODUCT CHALLENGES

- Who "owns" the trip data — one person or everyone?
- What happens when a member is added mid-trip?
- How to handle partial payers and unequal splits?
- How to make settlement feel final, not provisional?
- How to prevent the app from becoming Splitwise v2?

KEY TRADEOFF RESOLVED

The "who owns the trip" problem was resolved by making the first person to create the trip the implicit organizer — but allowing any member to log expenses. This mirrors how trips actually work socially: one person organizes, everyone contributes.

12 PRODUCT PSYCHOLOGY & BEHAVIORAL THINKING

Good products understand *how people actually behave* — not how they should.

TripKhata is built on several behavioral insights that shape how users interact with money, fairness, and social obligation.



Fairness Heuristic

People accept outcomes more readily when the process feels fair — even if amounts aren't perfect. Transparent timelines make the process visible, reducing dispute likelihood.



Loss Aversion

Framing debts as "X owes Y" triggers loss aversion. TripKhata frames it as "suggested transfers" — forward-looking, actionable, not accusatory. Language matters.



Closure Effect

Zeigarnik Effect: unresolved tasks occupy mental bandwidth. The "Mark Settled" button is a closure ritual — it tells the brain the loop is closed. Users report relief.



Transparency as Trust

The expense timeline serves a psychological function beyond utility — it creates a shared narrative of the trip. When everyone can see everything, suspicion dissolves.



Default Effects

Equal split is the default. Studies show defaults are followed 80%+ of the time. This one choice means most users never need to think about splitting.



Memory as Identity

The PDF report isn't just a receipt — it's a trip artifact. By framing the output as a "memory," users share it beyond necessity. This is how organic growth happens.

13 FUTURE ROADMAP

Where TripKhata goes *when the MVP is validated.*

The roadmap is sequenced by user value, not technical ambition. Phase 1 is live. Phases 2 and 3 are gated on real user feedback and retention data.

NOW MVP
Live

- Trip creation + member management
- Expense logging with auto equal split
- Minimum-transfer settlement algorithm
- Expense timeline view
- PDF trip report + WhatsApp sharing
- Personal expense tracking
- Shared trip wallet (deposit tracking)

NEXT v2.0

- User accounts + cloud sync (multi-device)
- Shareable trip link (friends join without install)
- Custom split (by amount or percentage)
- Expense categories with analytics
- WhatsApp Bot integration for logging via message

LATER
v3.0+

- UPI deep link integration ("Pay directly from app")
- AI expense categorization from description
- Trip templates (Shirdi trip, Goa, Trek, Wedding)
- Budget planning mode (pre-trip spend cap)
- Travel-brand partnerships + monetization layer

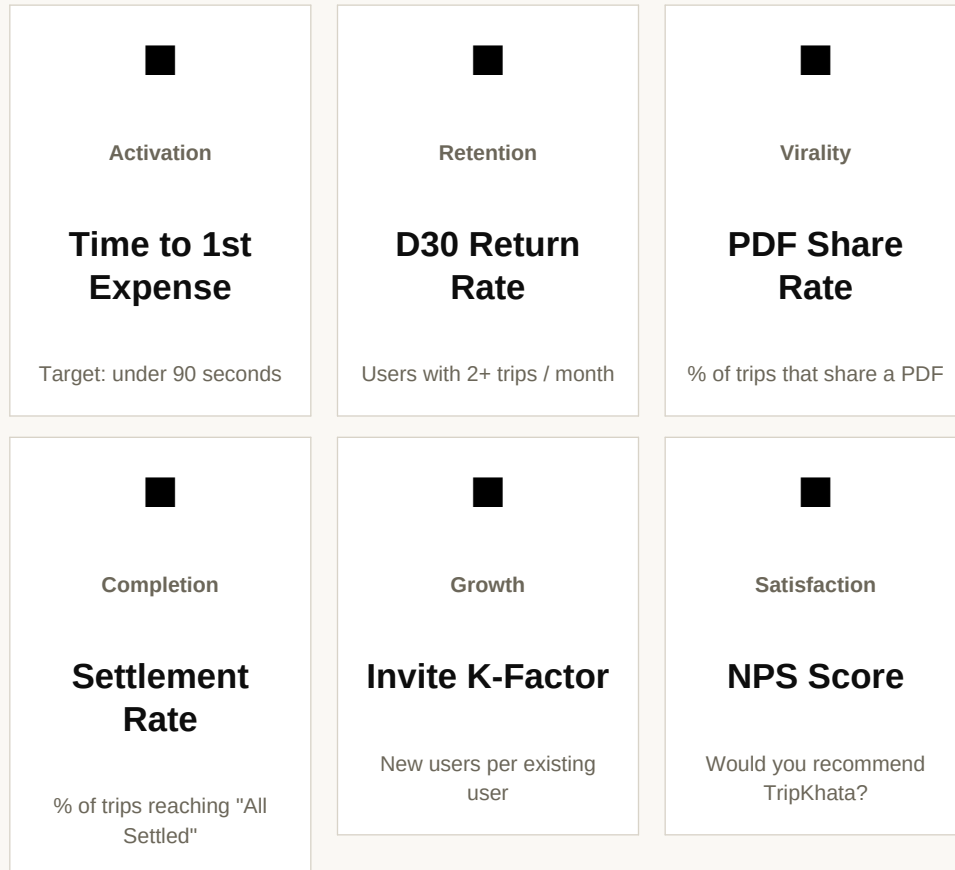
14 METRICS TO TRACK

What gets measured *gets built right.*

A product without metrics is a product without a compass. Below are the North Star metric and supporting metrics I would track across acquisition, engagement, and retention stages.

NORTH STAR METRIC

Trips Fully Settled Per Month — this captures the complete user journey (create → log → settle) and directly measures whether TripKhata is delivering its core promise.



ANTI-METRIC TO WATCH

Expenses per trip vs. member count ratio. If users are logging fewer expenses relative to trip size, it signals that the app is being abandoned mid-trip — the exact failure mode we're trying to prevent.

15 KEY PM LEARNINGS

What building TripKhata *taught me about product management.*

01

The best insight lives in the frustration, not the feature request

Users ask for "a better Notes app."
What they actually need is zero-friction expense memory.
Always ask why three times before building what users say they want.

02

Constraints force creativity

Building without a backend, without a team, and with a narrow scope taught me to be surgical. Every feature had to earn its place. That discipline produced a tighter product.

03

UX is product strategy

The decision to default to equal split is not just a design choice — it's a strategic choice about which user we serve. Every UX decision carries strategic weight.

04

Distribution is a feature

The PDF report was designed to be shareable on WhatsApp before it was designed to be functional. Virality must be built into the core experience, not added after launch.

05

Emotion drives engagement more than utility

Users don't just want to track money — they want to feel fair, feel trust, and feel closure. Products that solve emotional problems survive. Products that solve only functional problems get replaced.

06

Ship before perfect

The custom split feature wasn't built. The cloud sync wasn't built. But the core product works — and real users are using it. Shipped imperfect beats designed perfect every time.

16 FINAL REFLECTION

A product built with empathy, *shipped with conviction.*

TripKhata is, at its core, a product about relationships — not money. The money is just the surface. Underneath is the universal human desire to feel that things are fair, to travel without stress, and to remember a great trip without lingering financial awkwardness.

What I'm most proud of is not the algorithm or the UI — it's the discipline to say no to features that would have made TripKhata "more powerful" but less usable. Product management is, fundamentally, the art of making choices under uncertainty with limited resources on behalf of a user who can't always articulate what they need.

This case study is a snapshot of how I think about products. It represents not just TripKhata — but the PM I am becoming: empathy-first, data-informed, simplicity-obsessed, and founder-minded.

**Built with love and a real *Shirdi*
trip receipt in mind.**

Open to APM roles, startup product positions, and conversations with founders building for India.

Product Strategy · User Research · 0→1 Building · Prioritization · Behavioral Design ·
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